

CLAIMS

1. A therapeutic agent that inhibits poly [ADP-ribose] polymerase (“anti-PARP therapy”) for use in a method of treating a cancer patient, wherein the anti-PARP therapy is administered independent of DNA repair status, optionally independent of BRCA status, or
wherein the anti-PARP therapy is commenced prior to determining the DNA repair status of the patient or BRCA status of the patient, or
wherein the anti-PARP therapy is commenced in the absence of determining the DNA repair status of the patient or BRCA status of the patient.
2. The therapeutic agent for use according to claim 1, wherein the patient is characterized by an absence of a mutation in BRCA1 and/or BRCA2
3. A therapeutic agent that inhibits poly [ADP-ribose] polymerase (“anti-PARP therapy”) for use in a method of treating a cancer patient, wherein the patient is characterized by an absence of a mutation in BRCA1 and/or BRCA2, and wherein the cancer is a gynecological cancer selected from the group consisting of: ovarian cancer, fallopian tube cancer, peritoneal cancer, and breast cancer, or
wherein the cancer is characterized by an absence of a mutation in BRCA1 or BRCA2, and wherein the cancer is a gynecological cancer selected from the group consisting of: ovarian cancer, fallopian tube cancer, peritoneal cancer and breast cancer.
4. The therapeutic agent for use according to any one of claims 1-3, wherein the anti-PARP therapy is administered at a dose equivalent to about 100 mg, about 200 mg, or about 300 mg of niraparib or a salt or derivative thereof.
5. The therapeutic agent for use according to any one of claims 1-4, wherein the agent inhibits PARP-1 and/or PARP-2.
6. The therapeutic agent for use according to claim 5, wherein the agent is niraparib or a salt or derivative thereof.

7. The therapeutic agent for use according to claim 1 or claim 2, wherein the cancer is a gynecological cancer, optionally wherein the gynecological cancer is selected from the group consisting of: ovarian cancer, fallopian tube cancer, peritoneal cancer, and breast cancer.
8. The therapeutic agent for use according to any one of claims 1-7, wherein the cancer is a recurrent cancer.
9. The therapeutic agent for use according to any one of claims 1-8, wherein the method prolongs progression free survival as compared to control, and/or
wherein the method reduces the hazard ratio for disease progression or death as compared to control, and/or
wherein the method prolongs overall survival as compared to control, and/or
wherein the method achieves an overall response rate of at least 30%.
10. A therapeutic agent for use in a method of treating a patient having a recurrent cancer selected from ovarian cancer, fallopian tube cancer, or peritoneal cancer, wherein the therapeutic agent is niraparib, the method comprising administering niraparib to the patient according to a regimen determined to achieve i) prolonged progression free survival as compared to control, ii) a reduced hazard ratio for disease progression or death as compared to control, iii) prolonged overall survival as compared to control, or iv) an overall response rate of at least 30%.
11. The therapeutic agent for use according to any one of claims 1-10, wherein the cancer is ovarian cancer (e.g. wherein the patient has high grade serous ovarian cancer or high grade predominantly serous histology ovarian cancer), optionally wherein the ovarian cancer is platinum sensitive at the commencement of treatment, or
wherein the cancer is fallopian tube cancer, optionally wherein the fallopian tube cancer is platinum sensitive at the commencement of treatment, or
wherein the cancer is peritoneal cancer, optionally wherein the peritoneal cancer is platinum sensitive at the commencement of treatment.

12. The therapeutic agent for use according to any one of claims 1-11, wherein the patient is characterized by an absence of a mutation in BRCA1 and/or BRCA2, optionally wherein the patient is characterized by an absence of a germline mutation in BRCA1 and/or BRCA2, or wherein the patient is characterized by an absence of a sporadic mutation in BRCA1 and/or BRCA2.

13. The therapeutic agent for use according to any one of claims 1 and 4-11, wherein the patient has a tumor with a positive homologous recombination deficiency status.

14. The therapeutic agent for use according to any one of claims 1-12, wherein the patient has a tumor with a negative homologous recombination deficiency status.

15. The therapeutic agent for use according to any one of claims 10-14, wherein the regimen is determined to achieve a hazard ratio for disease progression of less than about 0.5, optionally wherein the hazard ratio for disease progression is less than about 0.3, and/or wherein the regimen is determined to achieve prolonged progression free survival of at least 9 months, optionally wherein the prolonged progression free survival is at least 12 months, especially wherein the prolonged progression free survival is at least 21 months.

16. The therapeutic agent for use according to any one of claims 1-15, wherein the method comprises continued treatment until disease progression or unacceptable toxicity, optionally wherein the method comprises reducing the dose of the anti-PARP therapy or niraparib in response to toxicity, especially wherein the method comprises reducing the dose of the anti-PARP therapy from about a dose equivalent to about 300 mg to a dose equivalent to about 200 mg, or wherein the method comprises reducing the dose of niraparib from about 300 mg to about 200 mg.

17. The therapeutic agent for use according to any one of claims 1-16, wherein the method comprises administering a plurality of oral doses, optionally wherein the method comprises once daily (QD) dosing, and/or

wherein the method comprises at least one 28 day cycle of dosing, and/or

wherein the oral dose is administered in one or more unit dosage forms, optionally wherein the one or more unit dosage forms are capsules.

18. The therapeutic agent for use according to any of claims 1-3, 5-15, and 17, wherein the oral dose is an amount within a range equivalent to about 5 to about 400 mg of niraparib, optionally wherein the amount is equivalent to about 5, about 10, about 25, about 50, about 100, about 150, about 200, about 250, about 300, about 350, or about 400 mg of niraparib,

especially wherein the amount is equivalent to about 300 mg of niraparib, or wherein the amount is equivalent to about 200 mg of niraparib, or wherein each unit dosage form comprises an amount equivalent to about 100 mg, about 200 mg, or about 300 mg of niraparib.

19. The therapeutic agent for use according to claim 17, wherein each QD dose is equivalent to about 100 mg, about 200 mg, or about 300 mg of niraparib, or

wherein each QD dose is administered as three unit dosage forms equivalent to about 100 mg of niraparib.

20. The therapeutic agent for use according to any one of claims 9-19, wherein progression free survival is characterized by a complete response of one or more target tumors, or

wherein progression free survival is characterized by a partial response of one or more target tumors.

21. The therapeutic agent for use according to any one of claims 1-20, wherein the method comprises administration to the patient in a fasted state, or

wherein the method comprises administration to the patient in a fed state.

22. The therapeutic agent for use according to any one of claims 9-21, wherein the prolonged progression free survival is at least 9 months, optionally wherein the prolonged progression free survival is at least 12 months, at least 15 months, at least 18 months, at least 21 months, at least 24 months, at least 27 months, at least 30 months, at least 33 months, or at least 36 months.

23. The therapeutic agent for use according to any one of claims 10-22, wherein the hazard ratio for disease progression is about 0.3, or

wherein the hazard ratio for disease progression is about 0.45, or

wherein the hazard ratio for disease progression is about 0.5, or

wherein the hazard ratio for disease progression is less than about 0.5, optionally wherein the hazard ratio for disease progression is less than about 0.45, less than about 0.4, less than about 0.35, or less than about 0.3.

24. The therapeutic agent for use according to any one of claims 1-23, wherein anti-PARP therapy or niraparib is administered as a maintenance therapy.

25. A therapeutic agent for use in a method of treating a patient lacking homologous repair deficiency and having a recurrent cancer selected from platinum-sensitive ovarian cancer, fallopian tube cancer, or primary peritoneal cancer, wherein the therapeutic agent is niraparib, the method comprising administering niraparib to the patient according to a regimen determined to achieve prolonged progression free survival as compared to control.

26. The therapeutic agent for use according to any one of claims 1-22, wherein the method has further been determined to achieve improved progression free survival 2 as compared to control.

27. The therapeutic agent for use according to any one of claims 1-26, wherein the method has further been determined to achieve improved chemotherapy free interval as compared to control, and/or

wherein the method has further been determined to achieve improved time to first subsequent therapy as compared to control, and/or

wherein the method has further been determined to achieve improved time to second subsequent therapy as compared to control, and/or

wherein the method has been determined to not have a detrimental effect on Quality of Life as determined by FOSI and/or EQ-5D-5L, and/or

wherein the method has been determined to not impact the effectiveness of a subsequent treatment with a chemotherapeutic agent, optionally wherein the next chemotherapeutic agent is a platinum agent, especially wherein the platinum agent is selected from cisplatin, carboplatin, oxaliplatin, nedaplatin, triplatin tetranitrate, phenanthriplatin, picoplatin, or satraplatin.